EVALUATION

I. Multiple choice questions

1. The barrier potential of a silicon diode is approximately,

a) 0.7 V b) 0.3V c) 2.0 V d) 2.2V

2. If a small amount of antimony (Sb) is added to germanium crystal, (AIPMT 2011)

a) it becomes a p-type semiconductor

b) the antimony becomes an acceptor atom

c) there will be more free electrons than hole in the semiconductor

d) its resistance is increased

3. In an unbiased *p-n* junction, the majority charge carriers (that is, holes) in the *p*-region diffuse into *n*-region because of

a) the potential difference across the *p-n* junction

b) the higher hole concentration in *p*-region than that in n-region

c) the attraction of free electrons of *n*-region

d) All of the above

4. If a positive half –wave rectified voltage is fed to a load resistor, for which part of a cycle there will be current flow through the load?

a) 0^0–90^0 b) 90^0–180^0

c) 0^0–180^0 d) 0^0–360^0

5. The zener diode is primarily used as

a) Rectifier b) Amplifier

c) Oscillator d) Voltage regulator

6. The principle based on which a solar cell operates is

a) Diffusion b) Recombination

c) Photovoltaic action

d) Carrier flow

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7. The light emitted in an LED is due to

a) Recombination of charge carriers

b) Reflection of light due to lens action

c) Amplification of light falling at the junction

d) Large current capacity.

8. The barrier potential of a *p-n* junction depends on i) type of semiconductor material ii) amount of doping iii) temperature. Which one of the following is correct? (NEET)

a) (i) and (ii) only

b) (ii) only

c) (ii) and (iii) only

d) (i) (ii) and (iii)

9. To obtain sustained oscillation in an oscillator,

a) Feedback should be positive

b) Feedback factor must be unity

c) Phase shift must be 0 or 2π

d) All the above

10. If the input to the NOT gate is A = 1011, its output is

a) 0100 b) 1000

c) 1100 d) 0011

11. Which one of the following represents forward bias diode? (NEET)

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